

**Question 47:**

- a) Create EMP table with ECODE(primary key), ENAME, DCODE, GRADE, BASIC & JN-DT as the columns.[except BASIC & JN-DT,all columns are of char type and size of Grade is 1.]  
b) Insert number of rows.

**Query :**

```
CREATE TABLE EMP(
    ECODE CHAR(5) PRIMARY KEY,
    ENAME CHAR(20),
    DCODE CHAR(5),
    GRADE CHAR(1),
    BASIC NUMBER(8,2),
    JN_DT DATE
);

INSERT ALL
    INTO EMP VALUES ('E001','RAHUL','D01','A',25000,TO_DATE('10-JAN-2020','DD-MON-YYYY'))
    INTO EMP VALUES ('E002','AMIT','D01','A',28000,TO_DATE('15-FEB-2021','DD-MON-YYYY'))
    INTO EMP VALUES ('E003','SUMAN','D01','B',22000,TO_DATE('20-MAR-2022','DD-MON-YYYY'))
    INTO EMP VALUES ('E004','RINA','D02','A',30000,TO_DATE('12-APR-2020','DD-MON-YYYY'))
    INTO EMP VALUES ('E005','ANITA','D02','B',24000,TO_DATE('22-MAY-2021','DD-MON-YYYY'))
    INTO EMP VALUES ('E006','ARUN','D02','B',26000,TO_DATE('11-JUN-2022','DD-MON-YYYY'))
    INTO EMP VALUES ('E007','JOY','D03','A',35000,TO_DATE('05-JUL-2019','DD-MON-YYYY'))
    INTO EMP VALUES ('E008','PRIYA','D03','C',20000,TO_DATE('16-AUG-2020','DD-MON-YYYY'))
    INTO EMP VALUES ('E009','SOURAV','D03','C',21000,TO_DATE('25-SEP-2021','DD-MON-YYYY'))
SELECT * FROM DUAL;
```

**Output :**

```
SQL> CREATE TABLE EMP(
2     ECODE CHAR(5) PRIMARY KEY,
3     ENAME CHAR(20),
4     DCODE CHAR(5),
5     GRADE CHAR(1),
6     BASIC NUMBER(8,2),
7     JN_DT DATE
8 );
```

Table created.

```
SQL> INSERT ALL
2     INTO EMP VALUES ('E001','RAHUL','D01','A',25000,TO_DATE('10-JAN-2020','DD-MON-YYYY'))
3     INTO EMP VALUES ('E002','AMIT','D01','A',28000,TO_DATE('15-FEB-2021','DD-MON-YYYY'))
4     INTO EMP VALUES ('E003','SUMAN','D01','B',22000,TO_DATE('20-MAR-2022','DD-MON-YYYY'))
5     INTO EMP VALUES ('E004','RINA','D02','A',30000,TO_DATE('12-APR-2020','DD-MON-YYYY'))
6     INTO EMP VALUES ('E005','ANITA','D02','B',24000,TO_DATE('22-MAY-2021','DD-MON-YYYY'))
7     INTO EMP VALUES ('E006','ARUN','D02','B',26000,TO_DATE('11-JUN-2022','DD-MON-YYYY'))
8     INTO EMP VALUES ('E007','JOY','D03','A',35000,TO_DATE('05-JUL-2019','DD-MON-YYYY'))
9     INTO EMP VALUES ('E008','PRIYA','D03','C',20000,TO_DATE('16-AUG-2020','DD-MON-YYYY'))
10    INTO EMP VALUES ('E009','SOURAV','D03','C',21000,TO_DATE('25-SEP-2021','DD-MON-YYYY'))
11    SELECT * FROM DUAL;
```

9 rows created.

**Question 48 : change the column heading as shown below, So that in subsequent SELECT statement newly set heading will be shown:**

**ECODE EMPLOYEE CODE**

**ENAME NAME**

**DCODE DEPT.CODE**

**JN-DT JOINING DATE**

**Query :**

```
COLUMN ECODE HEADING 'EMPLOYEE CODE';
```

```
COLUMN ENAME HEADING 'NAME';
```

```
COLUMN DCODE HEADING 'DEPT.CODE';
```

```
COLUMN JN_DT HEADING 'JOINING DATE';
```

```
COLUMN
```

**Output :**

```
SQL> COLUMN ECODE HEADING 'EMPLOYEE CODE';
```

```
SQL> COLUMN ENAME HEADING 'NAME';
```

```
SQL> COLUMN DCODE HEADING 'DEPT.CODE';
```

```
SQL> COLUMN JN_DT HEADING 'JOINING DATE';
```

```
SQL> COLUMN
```

```
COLUMN    JN_DT ON
```

```
HEADING    'JOINING DATE'
```

```
COLUMN    DCODE ON
```

```
HEADING    'DEPT.CODE'
```

```
COLUMN    ENAME ON
```

```
HEADING    'NAME'
```

```
COLUMN    ECODE ON
```

```
HEADING    'EMPLOYEE CODE'
```

**Question 49:** Set the format of columns as mentioned below, So that in subsequent SELECT statement ,values appear in the specified format:

**\*format of BASIC is such that a value of 7000 will be shown as7,000**

**\*Format of GRADE will be such that full column name appears in the display.**

**\*For JN-DT format is such that 01-JAN-00 will be shown as JANURY 01,2000**

**Query :**

```
COLUMN BASIC FORMAT 999,999.99;
```

```
COLUMN GRADE HEADING 'GRADE';
```

```
COLUMN JN_DT FORMAT A20;
```

```
SELECT ECODE,  
       ENAME,  
       DCODE,  
       GRADE,  
       BASIC,  
       TO_CHAR(JN_DT, 'MONTH DD,YYYY') JN_DT  
FROM EMP;
```

**Output :**

```
SQL> COLUMN BASIC FORMAT 999,999.99;
```

```
SQL> COLUMN GRADE HEADING 'GRADE';
```

```
SQL> COLUMN JN_DT FORMAT A20;
```

```
SQL> SELECT ECODE,  
2          ENAME,  
3          DCODE,  
4          GRADE,  
5          BASIC,  
6          TO_CHAR(JN_DT, 'MONTH DD,YYYY') JN_DT  
7 FROM EMP;
```

| EMPLO | NAME   | DEPT. | G | BASIC     | JOINING   | DATE    |
|-------|--------|-------|---|-----------|-----------|---------|
| E001  | RAHUL  | D01   | A | 25,000.00 | JANUARY   | 10,2020 |
| E002  | AMIT   | D01   | A | 28,000.00 | FEBRUARY  | 15,2021 |
| E003  | SUMAN  | D01   | B | 22,000.00 | MARCH     | 20,2022 |
| E004  | RINA   | D02   | A | 30,000.00 | APRIL     | 12,2020 |
| E005  | ANITA  | D02   | B | 24,000.00 | MAY       | 22,2021 |
| E006  | ARUN   | D02   | B | 26,000.00 | JUNE      | 11,2022 |
| E007  | JOY    | D03   | A | 35,000.00 | JULY      | 05,2019 |
| E008  | PRIYA  | D03   | C | 20,000.00 | AUGUST    | 16,2020 |
| E009  | SOURAV | D03   | C | 21,000.00 | SEPTEMBER | 25,2021 |

9 rows selected.

**Question 50:**

a) Show the display attributes of all the columns.

Query :

COLUMN;

Output :

```
SQL> COLUMN;
COLUMN      GRADE ON
HEADING     'GRADE'

COLUMN      BASIC ON
FORMAT      999,999.99

COLUMN      JN_DT ON
HEADING     'JOINING DATE'
FORMAT      A20

COLUMN      DCODE ON
HEADING     'DEPT.CODE'

COLUMN      ENAME ON
HEADING     'NAME'

COLUMN      ECODE ON
HEADING     'EMPLOYEE CODE'
```

b) Show the display attributes of particular column.

Query :

COLUMN JN\_DT;

Output :

```
SQL> COLUMN JN_DT;
COLUMN      JN_DT ON
HEADING     'JOINING DATE'
FORMAT      A20
```

c) Suppress the newly set attributes of JN-DT .Try a select statement.

Query :

```
COLUMN JN_DT OFF;
SELECT * FROM EMP;
```

Output :

```
SQL> COLUMN JN_DT OFF;
SQL> SELECT * FROM EMP;
```

| EMPLO | NAME   | DEPT. | G | BASIC     | JN_DT     |
|-------|--------|-------|---|-----------|-----------|
| E001  | RAHUL  | D01   | A | 25,000.00 | 10-JAN-20 |
| E002  | AMIT   | D01   | A | 28,000.00 | 15-FEB-21 |
| E003  | SUMAN  | D01   | B | 22,000.00 | 20-MAR-22 |
| E004  | RINA   | D02   | A | 30,000.00 | 12-APR-20 |
| E005  | ANITA  | D02   | B | 24,000.00 | 22-MAY-21 |
| E006  | ARUN   | D02   | B | 26,000.00 | 11-JUN-22 |
| E007  | JOY    | D03   | A | 35,000.00 | 05-JUL-19 |
| E008  | PRIYA  | D03   | C | 20,000.00 | 16-AUG-20 |
| E009  | SOURAV | D03   | C | 21,000.00 | 25-SEP-21 |

9 rows selected.

**d) Reset the newly set attributes of JN-DT**

**Query :**

```
COLUMN JN_DT CLEAR
```

```
COLUMN JN_DT
```

**Output :**

```
SQL> COLUMN JN_DT CLEAR
```

```
SQL> COLUMN JN_DT
```

```
SP2-0046: COLUMN 'JN_DT' not defined
```

```
SQL> █
```

**e) Reset the newly set attributes of all columns.**

**Query :**

```
CLEAR COLUMNS;
```

**Output :**

```
SQL> CLEAR COLUMNS;
```

```
columns cleared
```

```
SQL> █
```

**f) Shown the display attributes of all columns.**

**Query :**

```
COLUMN;
```

**Output :**

```
SQL> COLUMN;
```

```
SP2-0045: * no COLUMN defined
```

```
SQL> █
```

**Question 51:**

a) Show the records from EMP table in the ascending order of DCODE . DCODE value will be shown only for the first record of that department (same of DCODE is not repeated)

Query :

```
BREAK ON DCODE;
```

```
SELECT ECODE, ENAME, DCODE, GRADE, BASIC FROM EMP ORDER BY DCODE;
```

Output :

```
SQL> BREAK ON DCODE;
SQL> SELECT ECODE,
  2      ENAME,
  3      DCODE,
  4      GRADE,
  5      BASIC
  6 FROM EMP
  7 ORDER BY DCODE;
```

| ECODE | ENAME  | DCODE | G | BASIC |
|-------|--------|-------|---|-------|
| E001  | RAHUL  | D01   | A | 25000 |
| E002  | AMIT   |       | A | 28000 |
| E003  | SUMAN  |       | B | 22000 |
| E004  | RINA   | D02   | A | 30000 |
| E005  | ANITA  |       | B | 24000 |
| E006  | ARUN   |       | B | 26000 |
| E007  | JOY    | D03   | A | 35000 |
| E008  | PRIYA  |       | C | 20000 |
| E009  | SOURAV |       | C | 21000 |

9 rows selected.

b) Further take measures so that ,after displaying the records of a department it skips one line .

Query :

```
BREAK ON DCODE SKIP 1;
```

```
SELECT ECODE, ENAME, DCODE, GRADE, BASIC FROM EMP ORDER BY DCODE;
```

Output :

```
SQL> BREAK ON DCODE SKIP 1;
SQL> SELECT ECODE,
  2      ENAME,
  3      DCODE,
  4      GRADE,
  5      BASIC
  6 FROM EMP
  7 ORDER BY DCODE;
```

| ECODE | ENAME  | DCODE | G | BASIC |
|-------|--------|-------|---|-------|
| E001  | RAHUL  | D01   | A | 25000 |
| E002  | AMIT   |       | A | 28000 |
| E003  | SUMAN  |       | B | 22000 |
| E004  | RINA   | D02   | A | 30000 |
| E005  | ANITA  |       | B | 24000 |
| E006  | ARUN   |       | B | 26000 |
| E007  | JOY    | D03   | A | 35000 |
| E008  | PRIYA  |       | C | 20000 |
| E009  | SOURAV |       | C | 21000 |

9 rows selected.

c) Further take measures so that records are also ordered on the basis of GRADE within a department & same GRADE value is not repeated.

Query :

```
BREAK ON DCODE SKIP 1 ON GRADE;
SELECT ECODE,
       ENAME,
       DCODE,
       GRADE,
       BASIC
FROM EMP
ORDER BY DCODE, GRADE;
```

Output :

```
SQL> BREAK ON DCODE SKIP 1 ON GRADE;
SQL> SELECT ECODE,
2          ENAME,
3          DCODE,
4          GRADE,
5          BASIC
6 FROM EMP
7 ORDER BY DCODE, GRADE;
```

| ECODE | ENAME  | DCODE | G | BASIC |
|-------|--------|-------|---|-------|
| E001  | RAHUL  | D01   | A | 25000 |
| E002  | AMIT   |       |   | 28000 |
| E003  | SUMAN  |       | B | 22000 |
| E004  | RINA   | D02   | A | 30000 |
| E005  | ANITA  |       | B | 24000 |
| E006  | ARUN   |       |   | 26000 |
| E007  | JOY    | D03   | A | 35000 |
| E008  | PRIYA  |       | C | 20000 |
| E009  | SOURAV |       |   | 21000 |

9 rows selected.

d) Take measure so that at the end of each GRADE in a department it will show average Basic for that grade in that department. At the end of each department, it will show the average & total Basic for the department. At the end of all departments it will show the overall total basic & average basic.

Query :

```
BREAK ON DCODE SKIP 1 ON GRADE SKIP 1;
```

```
COMPUTE AVG OF BASIC ON GRADE;
```

```
COMPUTE AVG OF BASIC ON DCODE;
```

```
COMPUTE SUM OF BASIC ON DCODE;
```

```
COMPUTE AVG OF BASIC ON REPORT;
```

```
COMPUTE SUM OF BASIC ON REPORT;
```

```
SELECT ECODE,
       ENAME,
       DCODE,
```



```

        GRADE,
        BASIC
FROM EMP
ORDER BY DCODE, GRADE;

```

**Output :**

```

SQL> BREAK ON DCODE SKIP 1 ON GRADE SKIP 1;
SQL> COMPUTE AVG OF BASIC ON GRADE;
SQL> COMPUTE AVG OF BASIC ON DCODE;
SQL> COMPUTE SUM OF BASIC ON DCODE;
SQL> COMPUTE AVG OF BASIC ON REPORT;
SQL> COMPUTE SUM OF BASIC ON REPORT;
SQL> SELECT ECODE,
2          ENAME,
3          DCODE,
4          GRADE,
5          BASIC
6 FROM EMP
7 ORDER BY DCODE, GRADE;

```

| ECODE | ENAME | DCODE | G     | BASIC |
|-------|-------|-------|-------|-------|
| ----- |       |       |       |       |
| E001  | RAHUL | D01   | A     | 25000 |
| E002  | AMIT  |       |       | 28000 |
|       |       |       | *     | ----- |
|       |       |       | a     | 26500 |
|       |       |       |       |       |
| E003  | SUMAN |       | B     | 22000 |
|       |       |       | *     | ----- |
|       |       |       | a     | 22000 |
|       |       |       |       |       |
|       |       |       | ***** | ----- |
|       |       |       | sum   | 75000 |

| ECODE | ENAME | DCODE | G | BASIC |
|-------|-------|-------|---|-------|
| ----- |       |       |   |       |
| E004  | RINA  | D02   | A | 30000 |
|       |       |       | * | ----- |
|       |       |       | a | 30000 |
|       |       |       |   |       |
| E005  | ANITA |       | B | 24000 |
| E006  | ARUN  |       |   | 26000 |
|       |       |       | * | ----- |
|       |       |       | a | 25000 |

| ECODE | ENAME  | DCODE | G     | BASIC |
|-------|--------|-------|-------|-------|
| ----- |        |       |       |       |
|       |        |       | ***** | ----- |
|       |        |       | sum   | 80000 |
|       |        |       |       |       |
| E007  | JOY    | D03   | A     | 35000 |
|       |        |       | *     | ----- |
|       |        |       | a     | 35000 |
|       |        |       |       |       |
| E008  | PRIYA  |       | C     | 20000 |
| E009  | SOURAV |       |       | 21000 |
|       |        |       | *     | ----- |
|       |        |       | a     | 20500 |

| ECODE | ENAME | DCODE | G     | BASIC |
|-------|-------|-------|-------|-------|
| ----- |       |       |       |       |
|       |       |       | ***** | ----- |
|       |       |       | sum   | 76000 |

9 rows selected.